MIL-A-43316A(GL)

20 June 1967

SUPERSEDING
MIL-A-43316(GL)
15 March 1965

MILITARY SPECIFICATION

ADHESIVE, PATCHING FOR CHLOROPRENE COATED OR CHLOROSULPHONATED POLYETHYLENE COATED FABRICS

1. SCOPE

1.1 This specification covers requirements for one type of solvent adhesive for patching coated fabrics (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-T-191	- Textile Test Methods.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.

MILITARY

MIL-C-43285(GL)	- Cloth, Coated (Chloroprene Base Coated, Chlorosulphonated Polyethylene Top Coated).

STANDARDS

FEDERAL

Fed. Test Method Std. No. 175	- Adhesives: Methods of Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

FSC 8040

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF HEALTH EDUCATION AND WELFARE
Federal Hazardous Substances Labeling Act

(Application for copies should be addressed to the U.S. Department of Health, Education and Welfare, Food and Drug Administration, Washington, D. C. 20203.)

3. REQUIREMENTS

3.1 Composition.- The adhesive furnished under this specification shall be based on an oil-resistant elastomer in a solvent blend of petroleum naphtha, toluol, xylol or ketones.

3.2 Physical properties.-

3.2.1 Solids content.- The solids content shall range from 25-30 percent by weight, when tested as specified in 4.2.3.

3.2.2 Shear strength.- The shear strength shall be 100 pounds minimum when tested as specified in 4.2.3.

3.2.3 Resistance to dead load.-

3.2.3.1 Initial.- There shall be not more than 1/8 inch slippage of the seam when tested as specified in 4.2.3.

3.2.3.2 After 24 hours aging of seam.- There shall be not more than 1/8 inch slippage of the seam when tested as specified in 4.2.3.

3.2.3.3 At -40°F.- There shall be not more than 1/8 inch slippage of the seam when tested as specified in 4.2.3.

3.2.4 Peel strength (dead load).- The amount of peel back shall be not over 1/8 inch when tested as specified in 4.2.3.

3.2.4.1 Dead load peel strength at -40°F.- The amount of peel back shall be not over 6 inches when tested as specified in 4.2.3.

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3.2.5 Stability.- After being subjected to the aging condition specified in 4.3.6, the adhesive shall have a shear adhesion of 100 pounds minimum when tested as specified in 4.2.3.

3.2.5.1 Storage stability.- The manufacturer shall certify that he posesses storage data indicating that adhesive of the same composition as that being furnished has remained suitable for use after one year minimum storage in an ambient temperature range of 60 to 90°F.

3.3 Brushing consistency.- The adhesive shall be capable of being readily applied by brush when its temperature is 32°F. and higher, when tested as specified in 4.2.3.

3.4 Shear strength after freeze-thaw.- The adhesive shall have a shear adhesion of 100 lbs. minimum when tested as specified in 4.2.3.

3.5 Workmanship.- The adhesive shall be homogeneous, free from dirt and foreign matter, manufactured to conform to the established quality provisions of this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs) in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Sampling for inspection.- Sampling for inspection shall be performed in accordance with the provisions of MIL-STD-105 except where otherwise indicated herein.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

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4.2.2 Examination of the end item.-

4.2.2.1 Visual examination.- The packaged adhesive shall be examined for defects in appearance, fill, closure, condition of container and brushing consistency. The sample unit shall be one container of adhesive. The lot size for purpose of sampling shall be expressed in units of one container of adhesive.

Examine	Defect
Appearance	Not liquid. Not homogeneous. Any dirt or foreign matter.
Package (can)	Not type specified. Precautionary warning label missing. Any broken or damaged. Not outside coated as specified. Any leakage.
Contents of package (can)	Less than specified.

4.2.2.2 Examination of preparation for delivery.- An examination will be made to determine that markings, packaging and packing contents comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Markings (exterior and interior)	Incorrect, illegible, omitted; of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of component, such as incomplete closure of container flaps, improper taping, loose strapping, etc. Bulged or distorted container.
Content	Number per container is more or less than required.

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4.2.2.3 Inspection levels and acceptable quality levels (AQLs) for examination.- The inspection levels for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
For defects in 4.2.2.1	I	4.0
For defects in 4.2.2.2	S-2	2.5

4.2.3 Testing of the end item.- The end item adhesive shall be tested for the characteristics listed in table I. The lot shall consist of a manufacturer's batch (all material manufactured at the same time using the same raw materials) and shall be expressed in units of one pint can of adhesive each. The sample unit shall be three one pint cans of adhesive. The sample size shall be two. There shall be no evidence of failure of any sample unit to meet the requirements specified.

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INSTRUCTIONS FOR TESTING

TABLE I

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit 2/	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Composition	3.1	1/							
Solids content	3.2.1	4.3.1	X		2		per cent		
Shear strength	3.2.2	4.3.3	X		2	X			
Resistance (dead load)									
Initial	3.2.3.1	4.3.4	X		2	X			
After 24 hour aging	3.2.3.2	4.3.4.1	X		2	X			
At -40°F.	3.2.3.3	4.3.4.2	X		2	X			
Peel strength (dead load)	3.2.4	4.3.5.1.1	X		2	X			
At -40°F.	3.2.4.1	4.3.5.2.2	X		2	X			
Stability	3.2.5	4.3.6	X		2	X			
Brushing consistency	3.3	4.3.7	X		2	X			
Shear Strength after freeze-thaw	3.4	4.3.8	X		2	X			
Storage stability	3.2.5.1	1/							

1/ A certificate of compliance shall be furnished the contracting officer by the supplier.

2/ Both specimens shall pass the test.

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4.3 Test methods.-

4.3.1 Solids content.- The solids content shall be tested in accordance with Method 4021, Procedure A of Fed. Test Method Std. No. 175.

4.3.2 Preparation of specimens for shear and dead load resistance tests.- The specimens shall be 1 inch wide strips of coated fabric having an adhesive-bonded 2 inch overlap at the center. The coated fabric shall conform to MIL-C-43285(GL). The warp direction of the coated fabric shall be parallel to the long dimension of the specimen. The adhesive bond shall be made between the face side of one member of the seam and the back side of the other member, i.e., a neoprene coated side shall be bonded to a chlorosulphonated polyethylene top coated side. The strips shall extend approximately 8 inches beyond the overlapped area. The 1-inch wide strips shall be cut from a single large lap seam sample, (approximately 1 foot wide) prepared as follows. The surfaces to be bonded shall be abraded by hand using 80 grit abrasive cloth. The abrading shall be done uniformly (5 to 6 rubs per unit area) so that the entire area becomes roughened with a minimum of breaks through the coated surface. The surfaces shall be washed again with toluol and a clean wiping paper (see 6.3) and dried for 5 to 10 minutes, (see 6.3). Two coats of the adhesive to be tested shall be brushed on both surfaces and each coat shall be dried 5 minutes. The cement-coated surfaces shall be carefully brought together to form the required 2-inch overlap seam, with warp threads aligned. The seam shall be pressed with a hand steel roller. Make note of the clock time when the seam is formed, to time the intervals required prior to subsequent tests. Cut out 1 inch wide strips at right angles across the seam as required for the strength tests.

4.3.3 Shear strength.- The shear strength shall be tested in accordance with Method 5102 of CCC-T-191 using two 1 inch wide test specimens prepared as described in 4.3.2 and centering the lap seam of the specimen between the tester jaws. Conduct the test on specimens conditioned for 24 hours at room temperature after preparation.

4.3.4 Resistance to dead load (initial).- Two specimens prepared as described in 4.3.2 shall be tested, beginning the test not sooner than 10 minutes and not later than 20 minutes after combining. One end of each specimen shall be fastened in a horizontal clamp or equivalent support. Across the other end of each specimen, attach a clamp and weight, which, together, shall be two pounds, uniformly distributed and in such a manner that the specimen hangs freely without twist. Allow the weight to hang in this manner for four hours at room temperature (70° to 90°F), then measure amount of slippage of the bonded area.

4.3.4.1 Resistance to dead load (after 24 hours).- Test shall be conducted as specified in 4.3.4 except that a 30-pound weight (clamp and weight) shall

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be used and the test specimens shall be conditioned at room temperature (70° to 90°F) for 24 hours after combining and prior to testing. Measure amount of slippage of the bonded area.

4.3.4.2 Resistance to dead load (at -40°F).- The test shall be conducted as specified in 4.3.4.1 except that the specimens shall be subjected to the dead load at -40°F.

4.3.5 Peel strength.-

4.3.5.1 Specimen preparation (dead load).- Specimens shall be as specified in Method 1041.1 of Fed. Test Method Std. No. 175 using the same materials and adhesive application procedure as specified in 4.3.2 for shear test specimens. The warp of the fabric of the specimen shall be parallel to the long dimension of the specimen. The adhesive bond shall be between the face side of one component of the specimen and the back side of the other component. Allow the specimen to condition for 24 hours at room temperature (70° to 90°F).

4.3.5.1.1 Resistance to dead load peel.- Immediately after the 24 hour conditioning period, fasten one tab end of the specimen in a horizontal clamp. Across the other tab end, attach a horizontal clamp with weight. Clamp and weight together shall be 2 pounds and shall be attached in such a way that the specimen hangs freely without twist. Allow the weight to hang in this manner for 4 hours at room temperature, then measure the distance of any peeling that occurred in the bonded area during the 4 hour test.

4.3.5.2 Specimen preparation for dead load peel resistance at -40°F.- Two test specimens shall be prepared and made up as follows: Working at room temperature, cut two 1-inch x 10-inch strips and two 2-inch x 10-inch pieces from the coated fabric, with the long dimensions along the warp. Abrade the backs of the 1-inch x 10-inch strips and the faces of the 2-inch 2-inch x 10-inch pieces and wash and dry, as described in 4.3.2. Three coats of the adhesive shall be brush applied to the backs of the two abraded and washed 1-inch x 10-inch strips allowing ten minutes drying time between coats and after the third coat. The two 1-inch x 10-inch cement-coated strips and the two 2-inch x 10-inch abraded, washed and dried pieces shall be placed in the test chamber at -40°F. and allowed to condition for 20 minutes. At this point, place a 1-pint round can of the adhesive to be tested and an applicator brush and hand roller into the test chamber. One brush coat of cement shall be applied immediately to the faces of the 2-inch x 10-inch pieces. Immediately the pre-cemented strips shall be laminated to the wet cemented pieces along their centerlines, for a distance of 8 inches, leaving 2-inch long adhered tabs for clamping during subsequent load test.

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4.3.5.2.1 Resistance for dead load peel test at -40°F.- Bonded specimens prepared as in 4.3.5 shall be hung on a frame so as to permit the attachment of a weight to cause 180° peeling stress to be applied to the bond. The 2 inch wide tab end of each of the two bonded specimens shall be fastened to the uppermost horizontal bar of a metal frame. After 15 minutes, a grip shall be attached to each of the 1 inch wide tabs and a 4-oz. weight applied and allowed to hang freely to exert a 180° peel stress. A mark shall be made at the line where the fabrics join. The weight shall be allowed to hang 5 minutes at which time a measurement shall be made of the distance which the seam has separated (peeled), in inches.

4.3.6 Stability.- A one-pint can of the adhesive shall be kept in an air circulating oven at 120°F. ± 2°F. for two weeks. It shall then be removed from the oven and allowed to cool for 3 hours. The adhesive shall be tested for shear strength as specified in 4.3.3.

4.3.7 Brushing consistency.- An approx. 2-foot square sample of coated fabric conforming to MIL-C-43285(GL) shall be used in this test. The chloroprene coated side shall be washed and scrubbed clean with a toluene wetted wiping paper, and allowed to dry 5 minutes. The test adhesive shall be at a temperature of 75°F. plus or minus 5°F. Five brushfuls (2 1/2-inch wide wall brush) shall be transferred to the cleaned coated fabric, brushing out each time and noting whether the adhesive brushes easily, i.e. without drag, and whether it flows and spreads readily. Repeat this test after placing the can of adhesive in a chamber at -40°F., stirring occasionally, and allowing the contents to cool to 32°F. Proceed with the brushing immediately when the temperature of the adhesive reaches 32°F., i.e. before gelation can occur.

4.3.8 Freeze-thaw stability.- A pint can of adhesive shall be placed in a cold chamber and brought to -65°F. ± 2°F. After four hours, the can of adhesive shall be removed and allowed to thaw at room temperature (70°- 90°F) for four hours. The adhesive shall than be tested for shear strength in the same manner as specified in 4.3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Adhesive shall be furnished in 1-pint round cans with ends double-seamed on, conforming to type V, class 2 of PPP-C-96. Each can shall be exterior coated in accordance with Plan B of PPP-C-96.

5.1.2 Level C.- Adhesive shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

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5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- Adhesive, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-C-96.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 and with the Federal Hazardous Substance Labeling Act. Marking of interior packages shall be accomplished by printing or stamping on a paper label which shall be affixed to each can. Interior package marking shall include the contract or order number, date of manufacture and the manufacturer's name.

5.3.1 Precautionary marking.- The following precautionary marking shall be placed on each label:

CAUTION

Volatile. Flammable.
May be irritating to skin, eyes, and respiratory system.
Keep away from heat or flame.
Keep closed when not in use.
Use only with adequate ventilation.
If eye or skin contact occurs, wash affected area with large amounts of water.

6. NOTES

6.1 Intended use.- The adhesive is for use in repair and patching inflated dual-wall shelters and other end items that are also made of chloroprene base coated, chlorosulphonated polyethylene top-coated synthetic fabric. The adhesive is required to effect repairs in a wide temperature range, including -40°F. when applied in the specified manner.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Wiping paper.- Wiping paper used in 4.3.2 may be obtained from the procuring activity. Requests for this paper should reference Federal Stock Number (FSN) 7920-634-1745.

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Army - GL

Review activities:

Army - GL, MD, ME, MR

Preparing activity:

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Project No. 8040-A043

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Adhesive, Patching For Chloroprene Coated or Chlorosulphonated Polyethylene Coated Fabrics - MIL-A-43316A (GL)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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